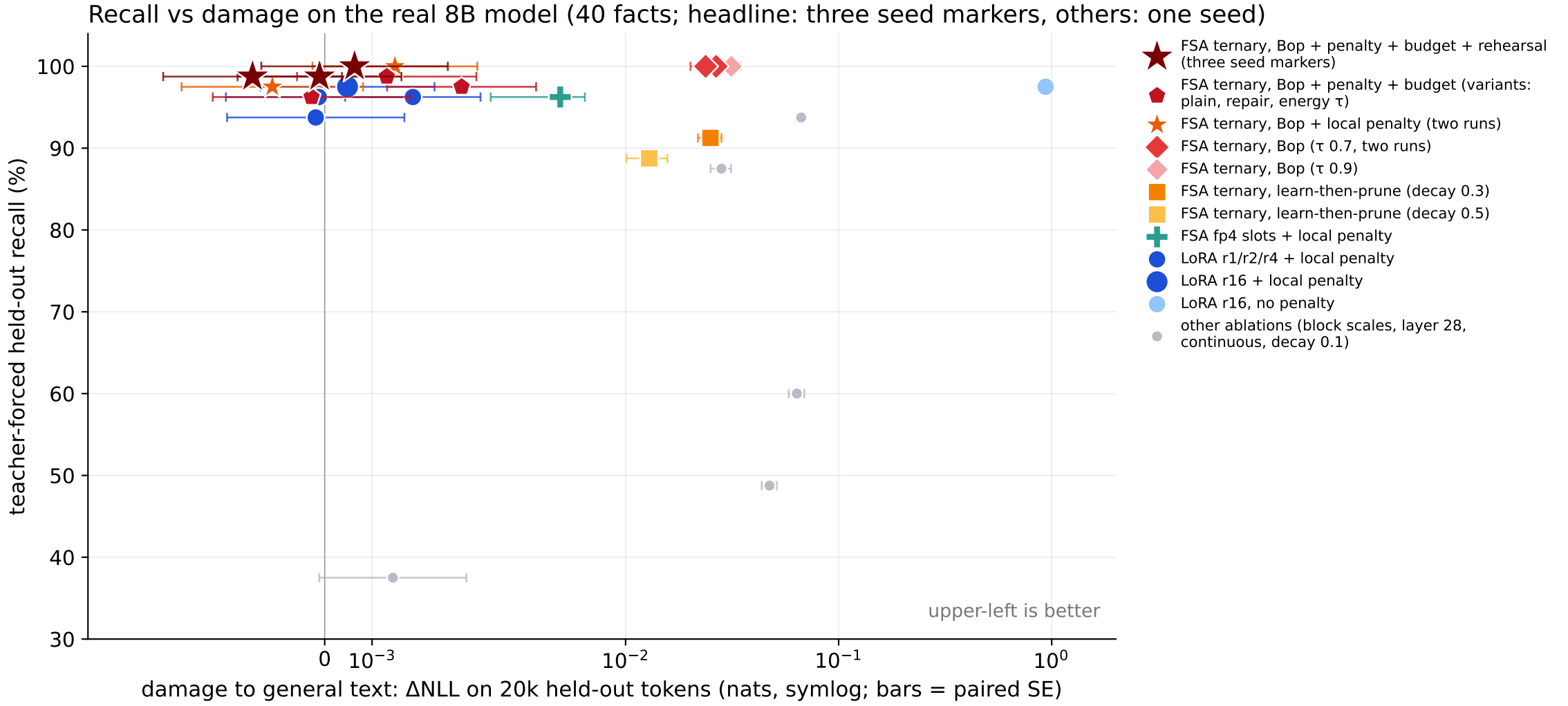
Recall vs damage on the real 8B model (40 facts; headline: three seed markers, others: one seed)
teacher-forced held-out recall (%)
100
90
80
70
60
50
40
30
0
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
damage to general text: ΔNLL on 20k held-out tokens (nats, symlog; bars = paired SE)
upper-left is better
FSA ternary, Bop + penalty + budget + rehearsal (three seed markers)
FSA ternary, Bop + penalty + budget (variants: plain, repair, energy τ)
FSA ternary, Bop + local penalty (two runs)
FSA ternary, Bop (τ 0.7, two runs)
FSA ternary, Bop (τ 0.9)
FSA ternary, learn-then-prune (decay 0.3)
FSA ternary, learn-then-prune (decay 0.5)
FSA fp4 slots + local penalty
LoRA r1/r2/r4 + local penalty
LoRA r16 + local penalty
LoRA r16, no penalty
other ablations (block scales, layer 28, continuous, decay 0.1)